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In [ ]: !jupyter nbconvert --to html d5_RNN_GRU_LSTM_Bidirectinality.ipynb
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In [4]: import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers
from datetime import datetime
from tensorflow.keras.datasets import mnist
```

```
In [5]: (x_train, y_train), (x_test, y_test) = mnist.load_data()
x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0
```

```
In [6]: #RNN - Recurrent Neural Network
model = keras.Sequential()
model.add(keras.Input(shape=(None, 28)))
model.add(
    layers.SimpleRNN(512, return_sequences=True, activation='relu')
)

model.add(layers.SimpleRNN(512, activation='relu'))
model.add(layers.Dense(10))

print(model.summary())

model.compile(
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    optimizer=keras.optimizers.Adam(learning_rate=0.001),
    metrics=["accuracy"]
)
print("Starting the function datetime:", datetime.now())
model.fit(x_train, y_train, batch_size=64, epochs=10, verbose=2)
model.evaluate(x_test, y_test, batch_size=64, verbose=2)
print("Ending the function datetime:", datetime.now())
```

Model: "sequential_1"

Layer (type)	Output Shape	Param #
simple_rnn_2 (SimpleRNN)	(None, None, 512)	276992
simple_rnn_3 (SimpleRNN)	(None, 512)	524800
dense_1 (Dense)	(None, 10)	5130

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Total params: 806922 (3.08 MB)
Trainable params: 806922 (3.08 MB)
Non-trainable params: 0 (0.00 Byte)

None

Starting the function datetime: 2025-07-31 15:35:09.517156

Epoch 1/10

938/938 - 179s - loss: 0.3082 - accuracy: 0.9053 - 179s/epoch - 191ms/step

Epoch 2/10

938/938 - 165s - loss: 0.1420 - accuracy: 0.9599 - 165s/epoch - 176ms/step

Epoch 3/10

938/938 - 167s - loss: 0.1167 - accuracy: 0.9678 - 167s/epoch - 179ms/step

Epoch 4/10

938/938 - 169s - loss: 0.1154 - accuracy: 0.9696 - 169s/epoch - 181ms/step

Epoch 5/10

938/938 - 180s - loss: 0.0953 - accuracy: 0.9740 - 180s/epoch - 192ms/step

Epoch 6/10

938/938 - 178s - loss: 0.0805 - accuracy: 0.9780 - 178s/epoch - 189ms/step

Epoch 7/10

938/938 - 189s - loss: 0.0886 - accuracy: 0.9762 - 189s/epoch - 201ms/step

Epoch 8/10

938/938 - 200s - loss: 0.0804 - accuracy: 0.9777 - 200s/epoch - 214ms/step

Epoch 9/10

938/938 - 174s - loss: 0.0712 - accuracy: 0.9803 - 174s/epoch - 186ms/step

Epoch 10/10

938/938 - 179s - loss: 0.0656 - accuracy: 0.9822 - 179s/epoch - 191ms/step

157/157 - 9s - loss: 0.0607 - accuracy: 0.9845 - 9s/epoch - 59ms/step

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In [7]: # GRU - Gated Recurrent Unit
model = keras.Sequential()
model.add(keras.Input(shape=(None, 28)))
model.add(
    layers.GRU(256, return_sequences=True, activation='tanh') # in activation use tanh
)

model.add(layers.GRU(256, activation='tanh')) # in activation use tanh
model.add(layers.Dense(10))

print(model.summary())

model.compile(
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    optimizer=keras.optimizers.Adam(learning_rate=0.001),
    metrics=["accuracy"]
)
```

```

)
print("Starting the function datetime:",datetime.now())
model.fit(x_train, y_train, batch_size=64, epochs=10, verbose=2)
model.evaluate(x_test, y_test, batch_size=64, verbose=2)
print("Ending the function datetime:",datetime.now())

```

Model: "sequential_2"

Layer (type)	Output Shape	Param #
=====		
gru (GRU)	(None, None, 256)	219648
gru_1 (GRU)	(None, 256)	394752
dense_2 (Dense)	(None, 10)	2570

```

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Total params: 616970 (2.35 MB)
Trainable params: 616970 (2.35 MB)
Non-trainable params: 0 (0.00 Byte)

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None

Starting the function datetime: 2025-07-31 16:10:32.983369

Epoch 1/10

938/938 - 128s - loss: 0.2746 - accuracy: 0.9086 - 128s/epoch - 136ms/step

Epoch 2/10

938/938 - 233s - loss: 0.0668 - accuracy: 0.9797 - 233s/epoch - 249ms/step

Epoch 3/10

938/938 - 233s - loss: 0.0473 - accuracy: 0.9855 - 233s/epoch - 249ms/step

Epoch 4/10

938/938 - 243s - loss: 0.0371 - accuracy: 0.9882 - 243s/epoch - 259ms/step

Epoch 5/10

938/938 - 237s - loss: 0.0301 - accuracy: 0.9907 - 237s/epoch - 252ms/step

Epoch 6/10

938/938 - 240s - loss: 0.0242 - accuracy: 0.9927 - 240s/epoch - 255ms/step

Epoch 7/10

938/938 - 244s - loss: 0.0229 - accuracy: 0.9927 - 244s/epoch - 261ms/step

Epoch 8/10

938/938 - 240s - loss: 0.0185 - accuracy: 0.9943 - 240s/epoch - 256ms/step

Epoch 9/10

938/938 - 220s - loss: 0.0160 - accuracy: 0.9948 - 220s/epoch - 234ms/step

Epoch 10/10

938/938 - 225s - loss: 0.0147 - accuracy: 0.9952 - 225s/epoch - 239ms/step

157/157 - 20s - loss: 0.0490 - accuracy: 0.9867 - 20s/epoch - 130ms/step

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In [8]: # LSTM - Long-short term memory
model= keras.Sequential()
model.add(keras.Input(shape=(None,28)))
model.add(
    layers.LSTM(256, return_sequences=True, activation='tanh')#in activation use ta
)

model.add(layers.LSTM(256, activation='tanh'))#in activation use tanh
model.add(layers.Dense(10))

```

```

print(model.summary())

model.compile(
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    optimizer=keras.optimizers.Adam(learning_rate=0.001),
    metrics=["accuracy"]
)
print("Starting the function datetime:",datetime.now())
model.fit(x_train, y_train, batch_size=64, epochs=10, verbose=2)
model.evaluate(x_test, y_test, batch_size=64, verbose=2)
print("Ending the function datetime:",datetime.now())

```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
=====		
lstm (LSTM)	(None, None, 256)	291840
lstm_1 (LSTM)	(None, 256)	525312
dense_3 (Dense)	(None, 10)	2570

=====

Total params: 819722 (3.13 MB)

Trainable params: 819722 (3.13 MB)

Non-trainable params: 0 (0.00 Byte)

None

Starting the function datetime: 2025-07-31 17:58:05.612160

Epoch 1/10

938/938 - 326s - loss: 0.3032 - accuracy: 0.9000 - 326s/epoch - 347ms/step

Epoch 2/10

938/938 - 270s - loss: 0.0849 - accuracy: 0.9738 - 270s/epoch - 288ms/step

Epoch 3/10

938/938 - 201s - loss: 0.0580 - accuracy: 0.9821 - 201s/epoch - 215ms/step

Epoch 4/10

938/938 - 179s - loss: 0.0444 - accuracy: 0.9863 - 179s/epoch - 191ms/step

Epoch 5/10

938/938 - 210s - loss: 0.0369 - accuracy: 0.9889 - 210s/epoch - 224ms/step

Epoch 6/10

938/938 - 256s - loss: 0.0315 - accuracy: 0.9904 - 256s/epoch - 273ms/step

Epoch 7/10

938/938 - 260s - loss: 0.0274 - accuracy: 0.9917 - 260s/epoch - 277ms/step

Epoch 8/10

938/938 - 205s - loss: 0.0240 - accuracy: 0.9921 - 205s/epoch - 218ms/step

Epoch 9/10

938/938 - 222s - loss: 0.0215 - accuracy: 0.9933 - 222s/epoch - 237ms/step

Epoch 10/10

938/938 - 190s - loss: 0.0201 - accuracy: 0.9936 - 190s/epoch - 203ms/step

157/157 - 16s - loss: 0.0321 - accuracy: 0.9896 - 16s/epoch - 100ms/step

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In []: